

PROJECT EXECUTABLE FILES

ComicCraft - AI Comic Story Creator using Gemini Models

Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	3 Marks

Team Members and Work Allocation

S.No	Team Member	Assigned Work
1	C. Iswarya	Project planning, requirements analysis, and overall project coordination
2	Hasini.D	UI/UX design and comic interface development
3	Ramya.A	Gemini API integration and AI story generation module
4	Nandhini	Comic panel generation, image handling, and content presentation
5	Nivetha.K	Testing, debugging, documentation, and final deployment support

Submission Checklist

S.No	Item to Submit	Status
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes / NA
4	Database schema / seed files (if applicable)	NA / If applicable
5	Environment configuration file (.env.example or similar)	Yes / If applicable
6	Deployed application URL (if hosted)	To be provided
7	APK / Executable binary (if applicable)	NA / If applicable
8	Dockerfile / Containerization config (if applicable)	NA / If applicable
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	To be provided

Step 2: File / Folder Structure

Provide the project's file and folder structure here. A typical structure may include the source code, UI components, Gemini integration module, generated comic assets, test files, configuration files, and README/setup documentation.

Suggested structure:

```
ComicCraft/  
■■■■ src/  
■■■■ components/  
■■■■ services/  
■■■■ assets/  
■■■■ tests/  
■■■■ .env.example  
■■■■ requirements.txt / package.json  
■■■■ README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	To be provided
Login Credentials (Demo)	To be provided, if applicable
Platform / Hosting Provider	To be provided
Repository Link	To be provided
Demo Video Link	To be provided

Step 4: Run Instructions

1. Download or clone the complete ComicCraft project files.
2. Install the required dependencies using the project's dependency file.
3. Configure the required Gemini API credentials in the environment configuration file.
4. Start the application using the project's configured run command.
5. Open the application in the provided local or deployed URL.
6. Enter a story idea or prompt and generate the AI-assisted comic story.
7. Verify the generated story, comic panels, and final presentation.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Gemini API availability or quota may affect AI generation.	Check API key, quota, and service availability.
2	Generated content may vary for the same prompt.	Use clear and detailed prompts.

S.No	Known Issue / Limitation	Workaround / Status
3	Image/content generation may take time depending on API response.	Wait for generation to complete and retry if needed.

Team Members

1. C. Iswarya	2. Hasini.D	3. Ramya.A
4. Nandhini	5. Nivetha.K	

PROJECT DOCUMENTATION

ComicCraft - AI Comic Story Creator using Gemini Models

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Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Team Members	C. Iswarya, Hasini.D, Ramya.A, Nandhini, Nivetha.K

Team Work Allocation

S.No	Name	Assigned Responsibility
1	C. Iswarya	Project planning, requirements analysis, overall coordination, and documentation
2	Hasini.D	UI/UX design, frontend interface, and user interaction flow
3	Ramya.A	Gemini model/API integration, prompt engineering, and AI story generation
4	Nandhini	Comic panel generation, image/content handling, and output presentation
5	Nivetha.K	Testing, debugging, validation, deployment support, and final verification

1. Project Description

ComicCraft is an AI-powered comic story creation platform designed to help users transform a simple idea or prompt into a structured comic story. The system uses Gemini Models to understand the user's concept, develop a storyline, create characters and dialogue, and organize the result into comic-style scenes or panels.

The application is intended to make comic creation easier for students, beginners, storytellers, and creative users by reducing the amount of manual planning required. Instead of starting with a blank page, the user can provide a theme, characters, setting, or short story idea and receive an AI-assisted comic concept.

Core Features

- AI-assisted story generation from a user-provided idea or prompt.
- Character and setting development based on the user's concept.
- Scene-by-scene comic structure with dialogue and narration.
- Gemini-based prompt processing and content generation.
- User-friendly interface for entering prompts and viewing generated results.
- Organized presentation of comic panels, captions, narration, and dialogue.

Project Objective

The main objective is to build a simple and creative AI application that combines generative AI with comic storytelling. ComicCraft should help users move from an initial idea to a readable comic story in a short and structured workflow.

2. Scenarios

Scenario 1 — Short Story to Comic

A user enters an idea such as a friendship adventure. ComicCraft develops the idea into a sequence of scenes with characters, narration, and dialogue suitable for a comic.

Scenario 2 — Character-Based Story

A user provides two or more characters and a setting. The system creates a connected storyline and assigns dialogue to the characters while maintaining the requested theme.

Scenario 3 — Educational Comic

A student provides an educational topic and asks for a simple comic explanation. The system organizes the topic into short scenes with easy-to-understand narration and dialogue.

3. Technical Architecture

ComicCraft can be organized as a modular application with three main layers: the user interface, application/backend logic, and Gemini AI integration. The frontend collects the user's story prompt and displays the generated comic content. The application layer validates input, builds structured prompts, and manages the generation flow. The Gemini layer processes the prompt and returns AI-generated story content.

Layer	Responsibility
Frontend / UI	Prompt input, options, loading state, comic result display, and user interaction
Application / Backend	Request handling, validation, prompt construction, response processing, and error handling
Gemini AI Integration	Story planning, character/dialogue generation, scene generation, and AI-assisted creative output
Output Layer	Structured comic panels, captions, narration, dialogue, and generated visual/content assets

Suggested Flow

User Prompt → Input Validation → Prompt Engineering → Gemini Model → Structured Story Response → Comic Scene/Panel Processing → Final Comic Display

4. Pre-requisites

- Basic knowledge of Python or the selected application programming language.
- HTML, CSS, and JavaScript knowledge if a web interface is used.
- Understanding of API requests and JSON data.
- Familiarity with Gemini API / Gemini Models and API key management.
- Basic prompt engineering concepts for generative AI.
- Git and version-control basics for collaborative development.
- A development environment capable of installing the project's required dependencies.

Security Note

The Gemini API key should be stored securely using environment variables or an equivalent secret-management method. It should not be hard-coded into source files or uploaded to a public repository.

5. Project Workflow

Milestone	Activities
Milestone 1: Planning & Model Selection	Define requirements, select Gemini model, decide application architecture, and set up the development environment.
Milestone 2: Story Generation	Design prompts and implement AI-assisted story, character, dialogue, and scene generation.
Milestone 3: Application Logic	Create request handling, input validation, response processing, and structured output logic.
Milestone 4: Frontend Development	Build the prompt interface, loading state, result cards, and comic-panel presentation.
Milestone 5: Testing & Deployment	Test different prompts, handle errors, verify output quality, and prepare the application for demonstration.
Milestone 6: Documentation & Conclusion	Prepare documentation, screenshots/demo materials, limitations, and future enhancements.

6. Milestone 1: Planning and Model Selection

Activity 1.1: Define Project Requirements

- Identify the target users and main comic-creation workflow.
- Define the minimum input required from the user.
- Decide the expected structure of generated stories and comic panels.
- List functional requirements such as story generation, dialogue generation, and result presentation.

Activity 1.2: Select Gemini Model

- Review available Gemini model capabilities and select a model appropriate for text/story generation.
- Consider response quality, context handling, speed, and project requirements.
- Configure the selected Gemini model through the supported API/SDK.

Activity 1.3: Define Architecture

- Define the interaction between frontend, backend/application logic, and Gemini.
- Decide how prompts and generated results are represented using structured data such as JSON.
- Plan error handling for invalid input, API failures, and incomplete responses.

Activity 1.4: Development Environment

- Create the project directory and virtual environment where applicable.
- Install the required SDKs and dependencies.
- Configure environment variables for the Gemini API key.
- Initialize version control and establish a shared team workflow.

7. Milestone 2: Core Story Generation

Activity 2.1: Prompt Engineering

- Create a system instruction that defines ComicCraft as a comic-story creation assistant.
- Pass the user's idea, genre, characters, setting, and other selected options into a structured prompt.
- Request a predictable output structure containing title, characters, scenes, narration, and dialogue.
- Keep generated content concise enough to be displayed clearly in comic panels.

Activity 2.2: Story Generation

- Send the structured prompt to the Gemini model.
- Receive the generated story response.
- Validate the response before displaying it.
- Convert the response into the application's comic-scene format.

Example Output Structure

```
Title
Character List
Scene 1: Setting + Narration + Dialogue
Scene 2: Setting + Narration + Dialogue
Scene 3: Setting + Narration + Dialogue
Final Scene / Ending
```

8. Milestone 3: Application Logic

Activity 3.1: Input Processing

- Receive the user's comic idea from the interface.
- Check that required fields are not empty.
- Limit excessively long input where necessary.
- Sanitize or validate input before building the AI request.

Activity 3.2: Gemini API Integration

- Load the Gemini API key securely from the environment.
- Initialize the Gemini client using the project's chosen SDK/API method.
- Send the structured prompt and receive the model response.
- Handle API errors, timeouts, and unexpected responses gracefully.

Activity 3.3: Response Processing

- Validate that the response contains the expected story sections.
- Extract scenes, dialogue, narration, and character information.
- Return structured data to the frontend.
- Show a useful error message if generation fails.

9. Milestone 4: Frontend Development

Activity 4.1: User Interface

- Create a clear landing/generator page for ComicCraft.
- Provide a text area or prompt field for the user's comic idea.
- Provide optional fields for genre, characters, setting, tone, or number of scenes where supported.
- Add a Generate Comic button with a visible loading state.

Activity 4.2: Comic Result Display

- Display the generated comic title prominently.
- Show characters in a separate section for easy reference.
- Present each scene as a comic panel/card.
- Separate narration/captions from character dialogue for readability.
- Provide a clear reset or generate-again action.

Activity 4.3: Responsive Design

- Ensure the layout works on desktop and mobile screens.
- Use consistent spacing, typography, and visual hierarchy.
- Avoid overcrowding long AI-generated responses.

10. Milestone 5: Testing and Deployment

Activity 5.1: Functional Testing

Test Case	Expected Result
Empty prompt	Application asks the user to provide a valid idea.
Normal story prompt	Comic story is generated and displayed in structured scenes.
Different genres	Generated story follows the selected genre/theme.
Multiple characters	Characters are used consistently in scenes and dialogue.
Long/invalid input	Application validates input and prevents problematic requests.
Gemini/API failure	Application displays a clear error and remains usable.

Activity 5.2: Deployment Preparation

- Install all dependencies listed by the project.
- Configure environment variables on the deployment system.
- Set the application to use production-safe settings where applicable.
- Verify the application after deployment using multiple sample prompts.

Activity 5.3: Demonstration

- Open the ComicCraft application.
- Enter a sample story idea.
- Generate the comic story.
- Show the title, characters, scenes, narration, and dialogue.

- Demonstrate another prompt to show that the system can create different stories.

11. Team Responsibilities

S.No	Name	Assigned Responsibility
1	C. Iswarya	Project planning, requirements analysis, overall coordination, and documentation
2	Hasini.D	UI/UX design, frontend interface, and user interaction flow
3	Ramya.A	Gemini model/API integration, prompt engineering, and AI story generation
4	Nandhini	Comic panel generation, image/content handling, and output presentation
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Collaboration Note

Although each member has a primary responsibility, the team should review and test the complete application together before submission. This helps ensure that the UI, Gemini integration, comic generation, testing, and documentation work as one system.

12. Exploring the Website / Application Pages

Home / Comic Generator Page

The main page acts as the entry point to ComicCraft. It provides the project title, a prompt input area, optional story controls, and the main Generate Comic action.

Input Section

The input section allows the user to describe the story idea. Optional selections such as genre, tone, characters, setting, and scene count can be included if implemented in the final application.

Generation State

After the user submits the prompt, the interface should show a loading indicator so that the user knows the AI generation request is being processed.

Results Section

The result area presents the generated comic title, character information, and individual scenes. Each scene can be represented as a panel containing narration, dialogue, and visual/content information.

13. Suggested File / Folder Structure

A suitable structure for a web-based implementation is shown below. The exact names may be adjusted to match the final codebase.

```
ComicCraft/  
■■■ app.py / main application file  
■■■ templates/  
■ ■■■ index.html  
■■■ static/  
■ ■■■ css/  
■ ■■■ js/  
■ ■■■ images/  
■■■ services/  
■ ■■■ gemini_service.py  
■■■ tests/  
■■■ .env.example  
■■■ requirements.txt / package file  
■■■ README.md
```

14. Run Instructions

1. Create or open the ComicCraft project directory.
2. Create and activate the project's virtual environment if the implementation uses Python.
3. Install the dependencies listed in the project dependency file.
4. Create an environment configuration file and add the Gemini API key securely.
5. Start the application using the project's configured run command.
6. Open the local application URL in a browser.
7. Enter a comic idea and generate the story.
8. Verify that the generated scenes, dialogue, narration, and other outputs are displayed correctly.

15. Known Issues / Limitations

S.No	Known Issue / Limitation	Suggested Workaround / Status
1	AI-generated content can vary between requests.	Use clear prompts and structured instructions.

S.No	Known Issue / Limitation	Suggested Workaround / Status
2	Gemini API availability or quota can affect generation.	Verify API key, quota, and service status.
3	Very long prompts can reduce readability or exceed practical limits.	Keep prompts focused and validate input length.
4	Generated visual content may require additional processing to be consistent with the final format and style.	Use a consistent panel format and validate assets before display.
5	AI output may require human review for accuracy and appropriateness.	Review generated content before final use.

16. Future Enhancements

- Add richer comic visual generation and consistent character appearance.
- Allow users to edit individual scenes or dialogue before finalizing the comic.
- Add downloadable comic formats such as PDF or image collections.
- Support multiple languages for story generation.
- Add user accounts and saved comic history.
- Provide additional genres, art styles, tones, and panel layouts.
- Improve character consistency across multiple generated scenes.

17. Conclusion

ComicCraft demonstrates how Generative AI can be applied to creative storytelling. By combining a simple user interface with Gemini Models, the project aims to transform short ideas into structured comic stories containing characters, scenes, narration, and dialogue.

The project covers model selection, prompt engineering, application development, frontend presentation, testing, and deployment preparation. Its modular structure also provides a foundation for future enhancements such as richer visual generation, scene editing, multilingual support, downloadable comics, and saved story history.

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